Course 6032A · Semester 6 · Program Elective · Revision 2021

Solar Power Technologies

A professional engineering handbook on solar thermal technologies, concentrated solar power, photovoltaic modules, rooftop solar arrays, PV system components, inverters, controllers, net metering and grid synchronization.

What this subject is

This course teaches how solar energy is converted into heat and electricity. It covers collectors, solar thermal devices, solar space heating/cooling, CSP plants, PV cells, rooftop arrays, solar module specifications, charge controllers, inverters, net meters and grid synchronization.

Malayalam: Solar technology ▯▯▯▯▯ ▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯: heat ▯▯▯▯▯▯▯▯▯ solar thermal systems; electricity ▯▯▯▯▯▯▯▯▯ solar PV systems. Rooftop solar-▯ module, inverter, controller, cable, protection, net meter ▯▯▯▯▯▯ ▯▯▯▯▯▯▯ select ▯▯▯▯▯▯▯.

Official syllabus information

Item	Value
Program	Diploma in Electrical and Electronics Engineering
Course code	6032A
Course title	Solar Power Technologies
Semester	6
Credits	4
Category	Program Elective
Periods	4 per week (L:3 T:1 P:0); 60 per semester

Course objectives

- Motivate students to adopt efficient solar technology as future energy techniques.
- Organize an outline for extracting electrical power or heat using solar technologies.
- Understand basic technology knowledge for solar PV system sizing and erection.

Prerequisite

Electrical Energy Generation, Transmission and Distribution, Semester 3.

Course outcomes

- CO1: Explain solar heat harnessing technologies.
- CO2: Classify concentrated solar power plants.
- CO3: Explain rooftop module array configurations and maintenance.
- CO4: Outline Solar PV operation and grid synchronization.

Redesigned syllabus roadmap

The official syllabus converted into a practical solar engineering learning route.

Module 1 · CO1 · 14 Hours

Solar Thermal Technologies

Scope of solar energy, solar collectors, solar thermal devices, industrial applications, space heating and solar absorption cooling.

Scope, advantages and limitations

Solar energy is clean, renewable and widely available. Its main advantages are low running cost, no fuel transport and reduced emissions. Limitations are intermittent availability, large area requirement, weather dependence and need for storage or backup.

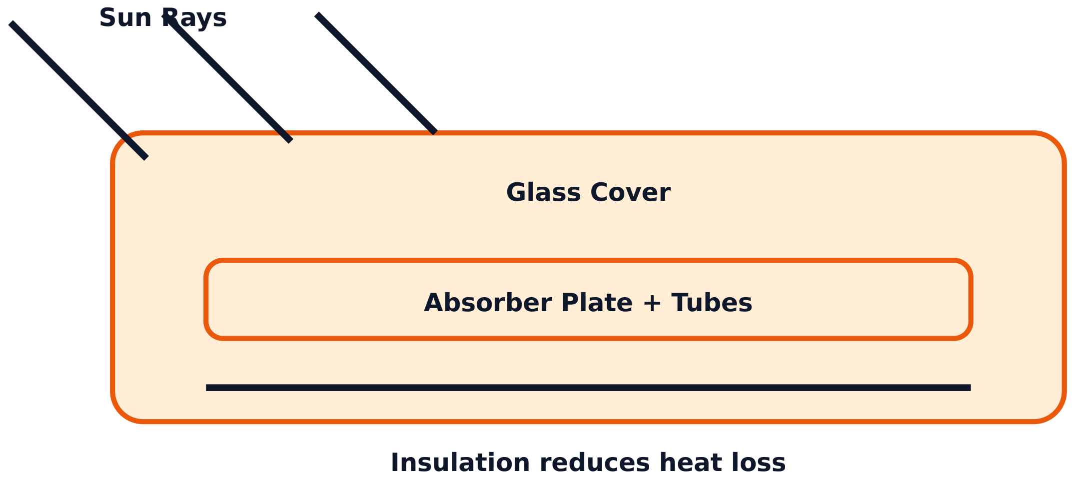
Malayalam: Solar system free fuel □□□□□□□□□□□□□□□□; □□□□□□ sunlight □□□□□□□□□ □□□□□ output □□□□□□□. □□□□□□□ battery, thermal storage, grid backup □□□□□□□□□□□□ hybrid arrangement □□□□□□□ □□□□.

Solar collectors

Solar collectors absorb radiation and transfer heat to a working fluid. Non-concentrating collectors collect both direct and diffuse radiation; concentrating collectors focus radiation to a smaller receiver and normally need tracking.

Type	Examples	Use
Non-concentrating	Flat plate, evacuated tube	Water heating, low temperature applications
Concentrating	Parabolic trough, mirror strip, Fresnel lens	Higher temperature heating and power generation

Flat plate collector concept



Solar thermal devices

Box type and dish type solar cookers convert solar radiation into cooking heat. Solar water heaters may use natural circulation or forced circulation. Solar stills evaporate and condense water. Solar air heaters, crop dryers, kilns, wax melters, pumps and greenhouses are industrial or agricultural applications.

Space heating and cooling

Passive solar heating uses building design: direct gain, indirect gain and isolated gain. Active solar heating uses hot air or steam heating systems with collectors, ducts or pumps. Solar absorption cooling uses heat energy to drive a refrigeration absorption cycle.

Expected questions

1. Explain advantages, disadvantages and applications of solar energy.
2. Compare concentrating and non-concentrating collectors.
3. Explain solar water heater with schematic diagram.
4. Explain passive and active solar space heating.
5. Draw block diagram of solar absorption cooling.

Module 2 · CO2 · 13 Hours

Concentrated Solar Power Plants

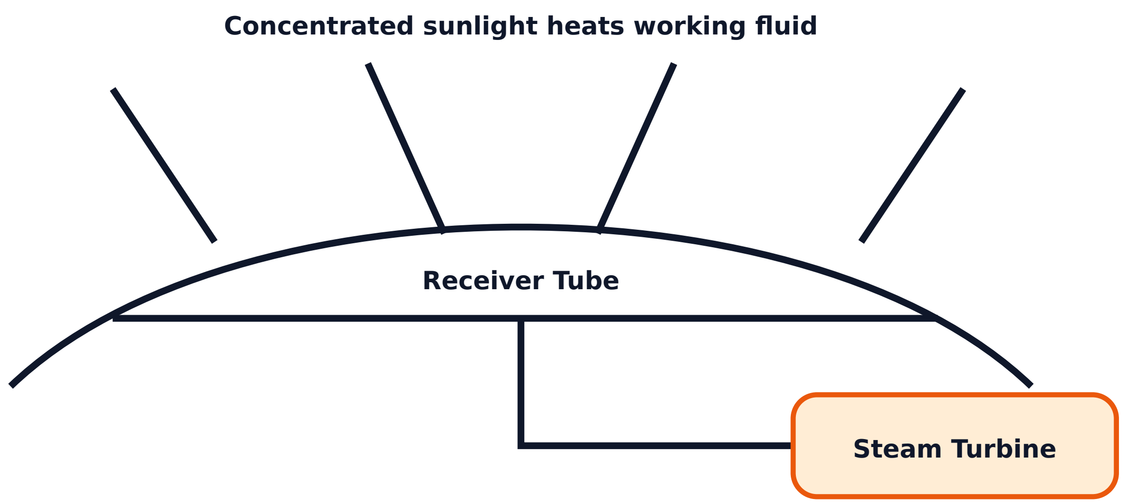
Classification of CSP plants based on temperature and working of low, medium and high temperature solar power plants.

CSP classification

Concentrated Solar Power systems use mirrors or lenses to concentrate sunlight and produce heat. The heat drives a thermodynamic cycle or engine. Based on temperature, plants may be low, medium or high temperature types.

- Low temperature: solar pond power plant and flat plate collector power plant.
- Medium temperature: line focusing parabolic collector power plant.
- High temperature: parabolic dish, central receiver, Fresnel reflector and Stirling engine systems.

Parabolic trough CSP



Low temperature plants

A solar pond works as a large heat collector and storage unit using salinity gradient to reduce convection. Heat from lower layers is used for low temperature power cycles. Flat plate collector power plants use multiple collectors to supply heat to a low temperature cycle.

Medium and high temperature plants

Line focusing parabolic collector plants focus sunlight onto a receiver tube. High temperature systems such as parabolic dish and central receiver power plants achieve high receiver temperatures and can drive Stirling engines or steam cycles.

Exam comparison

Plant	Focus type	Main feature
Parabolic trough	Line focus	Receiver tube carries heat transfer fluid
Parabolic dish	Point focus	High temperature, often Stirling engine
Central receiver	Heliostat field	Mirrors focus on tower receiver

Fresnel reflector	Segmented linear mirrors	Simpler mirror arrangement

Expected questions

1. Classify CSP plants based on temperature.
2. Explain solar pond power plant.
3. Explain parabolic trough power plant.
4. Compare parabolic dish and central receiver systems.

Module 3 · CO3 · 15 Hours

Solar PV Modules, Characteristics and Rooftop Arrays

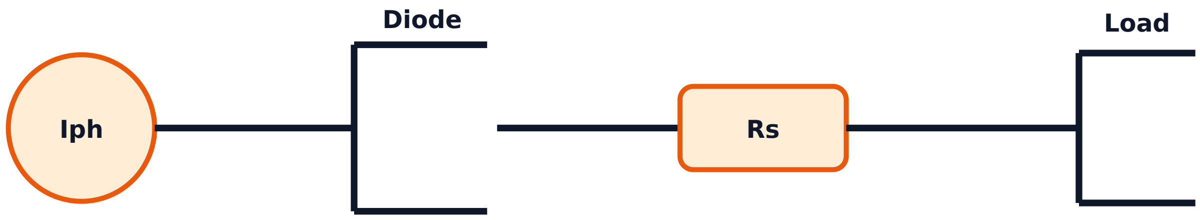
PV effect, solar cell construction, PV technologies, V-I characteristics, equivalent circuit, array configurations, specifications and maintenance.

Photovoltaic effect and PV technologies

A solar cell converts sunlight directly into DC electricity by photovoltaic effect. Light photons create electron-hole pairs in semiconductor material; the p-n junction separates charges and produces voltage. PV technologies include first-generation crystalline silicon, second-generation thin film and third-generation emerging technologies.

Malayalam: Solar cell sunlight directly DC electricity □□□□□□□□□□. Cell voltage/current □□□□□□□ cells series/parallel □□□ module, panel, array □□□□□□□□ connect □□□□□□□□□□.

Solar cell equivalent circuit



PV module characteristics

Important parameters are short-circuit current Isc, open-circuit voltage Voc, maximum power Pmax, voltage at maximum power Vmp, current at maximum power Imp and fill factor. The V-I curve shows how current and voltage vary with load.

Formula: Pmax = Vmp x Imp. Fill factor = Pmax / (Voc x Isc).

Array configuration

Definitions: cell, module, panel and array. Series connection increases voltage; parallel connection increases current. Rooftop systems may use single string series system, multiple string system or micro-inverter system.

- Single string: simple but affected by shading of one module.
- Multiple string: improves design flexibility.
- Micro-inverter: module level power conversion and better shade performance.

Specification and maintenance

Solar module specifications include electrical and mechanical parameters, STC ratings, module energy output, cell temperature, irradiance, air mass and derating factor. Installation factors include direction, tilt, shading and temperature. Preventive maintenance includes cleaning, visual inspection, loose connection check and output monitoring.

Expected questions

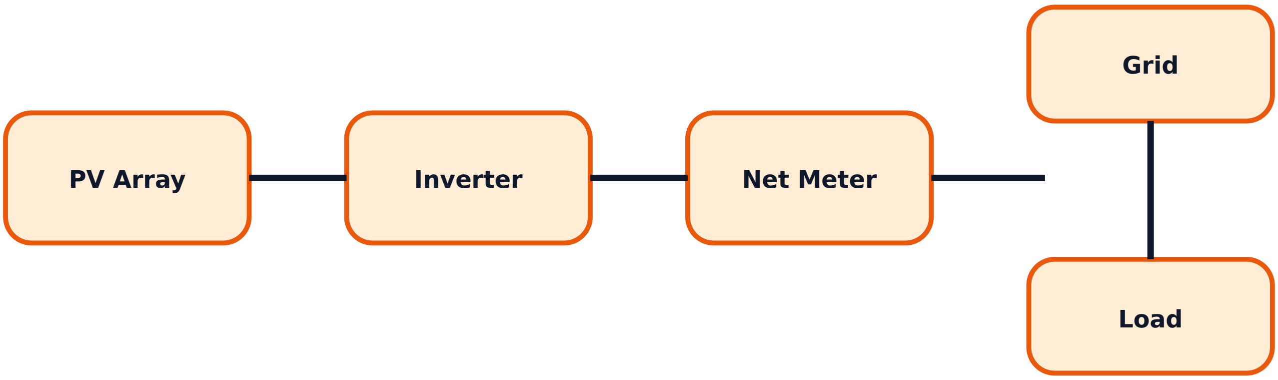
1. Explain photovoltaic effect with solar cell construction.
2. Draw and explain V-I characteristics of PV module.
3. Draw equivalent circuit of solar cell.
4. Explain series and parallel array configuration.
5. List factors affecting solar module output.

Module 4 · CO4 · 16 Hours

Solar PV System Operation and Grid Synchronization

PV system components, inverters, converters, charge controllers, MPPT/PWM, off-grid/on-grid/hybrid systems, net metering and grid synchronization.

Grid-tied PV system block



PV system components

PV system components include PV modules, storage battery, inverter, charge controller, solar array mounting structure, cables and connectors, blocking diode, bypass diode and junction box. Each component must be rated for voltage, current, temperature and outdoor conditions.

Inverters and controllers

Solar inverters convert DC output to AC. Specifications include DC input voltage/current range, AC voltage/frequency, efficiency, protection, enclosure and mechanical rating. Power converters such as buck and boost converters regulate voltage. Charge controllers may use PWM or MPPT.

> **Industrial note:** MPPT controller tracks maximum power point and is better than simple PWM when module voltage and battery/load voltage vary.

Types of PV systems

- **Off-grid:** battery based standalone system for isolated loads.
- **Grid-tied:** synchronized with grid, exports active power through net meter.
- **Hybrid:** combines solar, battery, grid and sometimes wind or generator.
- **Solar-wind hybrid:** uses complementary renewable sources.

Net metering and grid synchronization

Net meter records import and export energy. Active power export occurs when PV generation exceeds local load. Grid synchronization means inverter output voltage, frequency, phase and protection must match grid requirements before parallel operation.

Expected questions

1. List PV system components and functions.
2. Classify solar inverters and write specifications.
3. Explain buck, boost, charge controller, MPPT and PWM.

4. Draw block diagrams of off-grid, grid-tied and hybrid PV systems.
5. Explain net metering and grid synchronization.

Formula / Rule Bank

Solar PV formulas, sizing rules and diagram checklists.

PV power

P = V x I

Use for module output and load power.

Maximum power

Pmax = Vmp x Imp

Fill factor

FF = Pmax / (Voc x Isc)

Series modules

String voltage = number of modules in series x module voltage.

Parallel strings

Array current = number of parallel strings x string current.

Energy output

Energy = Power x sunshine hours x derating factor.

Tilt and direction rule

Install with correct direction, tilt, minimum shading and good ventilation.

Grid sync rule

Match voltage, frequency, phase sequence and protection before grid connection.

Solved Examples / Problem Lab

Step-by-step examples for PV calculations and exam answers.

Example 1 - Module power

Given: Vmp = 36 V, Imp = 8 A.

Pmax = Vmp x Imp = 36 x 8.

Pmax = 288 W.

Example 2 - Fill factor

Given: Voc = 45 V, Isc = 8.5 A, Pmax = 300 W.

FF = 300 / (45 x 8.5) = 300 / 382.5.

FF = 0.784.

Example 3 - String voltage

10 modules each with Vmp = 36 V are connected in series.

String Vmp = 10 x 36 = 360 V.

Example 4 - Array current

3 parallel strings, each string current = 8 A.

Array current = 3 x 8 = 24 A.

Example 5 - Daily energy

2 kW PV array, 5 sunshine hours, derating = 0.75.

Energy = 2 x 5 x 0.75.

Daily energy = 7.5 kWh.

Practice Section and Expected Questions

Module-wise exam preparation.

One-mark questions

1. What is photovoltaic effect?
2. Define Isc.
3. Define Voc.
4. What is MPPT?
5. What is net metering?

Short-answer questions

1. List applications of solar energy.
2. Compare flat plate and evacuated tube collectors.
3. List types of CSP plants.
4. List PV system components.
5. List factors affecting module output.

Descriptive questions

1. Explain solar thermal devices with examples.
2. Explain parabolic trough CSP plant with diagram.
3. Explain PV cell equivalent circuit and V-I curve.
4. Explain off-grid, grid-tied and hybrid PV systems.

MCQ practice

1. In series connection of PV modules, mainly increases: A) Current B) Voltage C) Frequency D) Shading

2. MPPT stands for: A) Maximum Power Point Tracking B) Minimum Power Panel Test C) Module Parallel Power Transfer D) Maximum Panel Protection Test

3. Net meter measures: A) Only import B) Only export C) Import and export D) Temperature

Fresh Model Question Paper

Original question paper based on syllabus coverage; not copied from official question papers.

Course 6032A - Solar Power Technologies

Time: 3 Hours   **Maximum Marks:** 100

Part A - Short Answer

1. Define solar collector.
2. What is photovoltaic effect?
3. Define fill factor.
4. What is bypass diode?
5. What is grid synchronization?

Part B - Medium Answer

1. Compare concentrating and non-concentrating collectors.
2. Explain solar water heater with diagram.
3. Classify CSP plants based on temperature.
4. Explain V-I characteristics of PV module.
5. Explain charge controller and MPPT.

Part C - Long Answer / Problems

1. Explain solar thermal devices used in domestic and industrial applications.
2. Draw and explain parabolic trough and central receiver CSP plants.
3. A module has Voc=44 V, Isc=9 A, Vmp=36 V and Imp=8 A. Calculate Pmax and fill factor.
4. Draw and explain grid-tied PV system with net metering.
5. Explain rooftop solar array configurations, installation factors and maintenance.

Complete Model Answers

Answer guide with formulas, diagram points and common mistakes.

Photovoltaic effect

Photovoltaic effect is the direct conversion of light energy into electrical energy in a semiconductor p-n junction when photons create electron-hole pairs.

PV problem answer

Pmax = Vmp x Imp = 36 x 8 = 288 W. Fill factor = Pmax / (Voc x Isc) = 288 / (44 x 9) = 288 / 396 = 0.727.

Pmax = 288 W, FF = 0.727.

Grid-tied PV diagram points

Draw PV array, DC isolator, inverter, AC protection, net meter, grid and load. Explain DC generation, inverter conversion, synchronization and active power export.

Rooftop maintenance points

Clean module glass, check shading, check cable/connector tightness, inspect mounting structure, inspect junction box, monitor output and verify inverter alarms.

Common mistakes

Do not mix up series and parallel connection. Series increases voltage; parallel increases current. Do not explain MPPT as only a protection device; it tracks maximum power point.

References from syllabus

Deambi, Chetan Singh Solanki, B.H. Khan, D.P. Kothari, David Buchla, O.P. Gupta, G.D. Rai, R.K. Rajaput, Swayam, NPTEL, TERI, MNRE and ANERT.